

Rimbun Logistics: FleetSense Telematics and Cold-Chain Platform

Detailed Design

Architecture, requirements, and validation criteria for build

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Revision history

Version	Date	Changes
1.0	2026-09-18	First consolidated detailed design.
1.1	2026-10-02	Post-review revisions: FR-11; backfill upload format and sequence (10); immobilisation procedure (13.3); alert service state and offsets (17.1); corrections and audit pack source (18.2, 18.3, 20.1); device authorisation policy (22.2); cellular data and cost budget (24.2, 24.3); NFR-4 acceptance criterion (28.2); Phase 1 scope wording (30); companion documents.

Changes since version 1.0

This version updates the sections listed below after the design review of 2026-09-25. Sections not listed are unchanged from version 1.0.

- FR-11 and Section 13.3: immobilisation procedure and approval.
- Section 10: format, size and sequence of backfill uploads from reefer gateways.
- Section 17.1: how the alert service keeps excursion state and Kafka offsets.
- Sections 18.2, 18.3 and 20.1: how corrections are recorded and which store the audit pack is generated from.
- Section 22.2: device authorisation policy and the source of the device identity.
- Sections 24.2 and 24.3: cellular plan, data volume and monthly run cost.
- Section 28.2: acceptance criterion for NFR-4.
- Section 30: Phase 1 scope wording.

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1. Purpose and Scope

This document describes the architecture of FleetSense, the telematics and cold-chain monitoring platform of Rimbun Logistics Berhad. FleetSense collects position, vehicle and driving data from every Rimbun truck and temperature data from every refrigerated trailer, turns that data into live operational views for dispatchers, alerts people when a temperature-controlled load is at risk, scores driving behaviour, updates gateway firmware over the air, and produces the cold-chain records that customers and regulators inspect. It is written to a level of detail sufficient for an engineering team to begin implementation, and Section 29 assesses where that is and is not yet true.

FleetSense replaces three systems: a vendor telematics portal used for the dry fleet in Malaysia and Singapore, a separate data-logger service used for reefer trailers, whose loggers are downloaded by cable at the depot, and a set of spreadsheets that the quality team uses to assemble temperature evidence for pharmaceutical customers. None of these share vehicle identities, and the reefer loggers give no live view at all.

In scope

Device connectivity and ingestion, offline buffering, the streaming backbone, live position and geofencing, driver behaviour events and scores, remote commands, over-the-air (OTA) firmware updates, cold-chain excursion detection and alerting, the quality review workflow and audit pack, the dispatcher console, the customer portal, consignee tracking links, storage, retention and the security model, for the fleet in Section 4.

Out of scope

Transport management (orders, route planning, billing), which remains in the existing TMS and is integrated through the trip feed in Section 6; workshop and maintenance management; the driver mobile app (a separate design consumes the FleetSense API); and fuel-card reconciliation.

Volume baseline

Metric	2026 actual	Design target (2028)
Trucks (prime movers and rigid) with a gateway	5,100	6,400
Reefer trailers with a gateway	1,800	2,300
Total connected gateways	6,900	8,700
Average driving hours per truck per day	10.4	11
Cold-chain trips per month	21,000	29,000
Share of cold-chain trips carrying pharmaceutical product	31%	38%
Customer and depot geofences	24,000	38,000

2. Requirements

Requirements are numbered for traceability to the acceptance criteria in Section 28.

2.1 Functional Requirements

ID	Requirement
FR-1	Ingest position and vehicle data from every truck gateway: while the truck is moving, a drive message every 10 seconds carrying ten 1 Hz frames; while the ignition is on and the truck is stationary, a position every 30 seconds; while the ignition is off, a position every 15 minutes.
FR-2	Ingest a cold-chain reading every 60 seconds from every reefer gateway: four load-space probes, return-air and supply-air temperature, setpoint, door state and reefer unit status.
FR-3	Buffer readings on the gateway for up to 72 hours without connectivity and deliver them to the platform on reconnection.
FR-4	Show every vehicle's live position, status and current trip on the dispatcher console map.
FR-5	Raise entry and exit events for depot, customer-site and restricted-zone geofences.
FR-6	Detect cold-chain excursions using the rules in Section 16.
FR-7	Alert an excursion to the dispatcher assigned to the trip and to the driver within 2 minutes of detection.
FR-8	Detect driver behaviour events (harsh braking, harsh acceleration, harsh cornering, speeding and excessive idling) and compute a daily driver score.
FR-9	Publish a monthly driver score to the HR system for the safe-driving incentive.
FR-10	Update gateway firmware over the air in waves, with automatic rollback.
FR-11	Send remote commands to gateways: reefer setpoint change, trailer door lock and unlock, and vehicle immobilisation (restart prevention).
FR-12	Give customers a portal with trip history, temperature charts, excursion summaries and proof of delivery.
FR-13	Give consignees a tracking link that shows the live position and estimated arrival of their delivery.
FR-14	Produce an audit pack per shipment: the continuous temperature record, excursions and their dispositions, mean kinetic temperature, any corrections, and the chain of custody.
FR-15	Provide a quality review workflow in which a quality officer reviews each excursion and records a disposition.
FR-16	Apply the retention and deletion rules in Section 21.

2.2 Non-Functional Requirements

ID	Requirement
NFR-1	Ingestion availability of 99.9% per calendar month, measured as the share of minutes in which the platform accepts device messages.
NFR-2	Completeness: at least 99.99% of the cold-chain readings taken by reefer gateways reach the cold-chain store, including after connectivity gaps of up to 72 hours.
NFR-3	Alert latency: 99% of excursion alerts are delivered within 2 minutes of the reading that completes the excursion rule reaching the platform.
NFR-4	Position freshness: for 99% of moving vehicles, the position shown on the dispatcher console is no more than 30 seconds old.
NFR-5	Record integrity: cold-chain records are complete, attributable and unaltered from capture to audit; an audit pack can be regenerated identically at any time within the retention period.
NFR-6	Device security: every gateway has a unique X.509 identity and all device traffic uses TLS 1.2 or later with mutual authentication.
NFR-7	Privacy: driver personal data is processed in line with the personal data protection laws of Malaysia, Singapore, Thailand and Indonesia.
NFR-8	Retention as set out in Section 21.
NFR-9	Recovery: RPO of zero on loss of an Availability Zone; on loss of the region, RPO of 5 minutes for operational data and no permanent loss of cold-chain readings; RTO of 4 hours.
NFR-10	Run cost no more than USD 95,000 per month at the 2028 fleet size, including cellular data.
NFR-11	Scale to 12,000 gateways without architectural change.
NFR-12	The dispatcher console map loads within 3 seconds for a dispatcher's full vehicle list (up to 600 vehicles).

3. Foundational Principles

#	Principle	Implication
P1	The gateway is the source of truth for what happened on the road.	Readings and events are timestamped and sequenced on the device; the platform never invents or interpolates a reading.
P2	At-least-once everywhere, idempotent consumers.	Every message may arrive more than once; every consumer deduplicates on the envelope key in Section 9.
P3	Cold-chain records are immutable from capture to audit.	A reading, once stored, is never altered; anything added later is recorded alongside it with who, when and why.
P4	Safety first.	The platform never degrades the driver's control of the vehicle.
P5	Least privilege for people, services and devices.	Each identity can reach only the data and actions its job needs.
P6	Driver location serves fleet operation only.	Driver location is visible only to Rimbun staff with an operational need and, while a delivery is under way, to the receiving customer for that delivery.
P7	Managed services before self-managed ones.	The team operates no broker, database or stream cluster it could buy as a managed service in the region.
P8	Design for poor coverage.	Rural routes in Malaysia, Thailand and Indonesia lose coverage for hours; every device-facing flow tolerates long disconnection.

4. Markets, Fleet and Customers

Rimbun operates line-haul and distribution in Malaysia (about 58% of trips), Singapore (11%), Thailand (18%) and Indonesia (13%), with cross-border line-haul between Malaysia and both Singapore and southern Thailand. Depots are in Shah Alam, Johor Bahru, Penang, Kuantan, Kuching, Kota Kinabalu, Singapore (Tuas), Bangkok (Lat Krabang), Hat Yai, Jakarta (Cikarang) and Surabaya.

The truck fleet is mixed: about 70% prime movers pulling Rimbun or customer trailers and 30% rigid trucks used for urban distribution. Reefer trailers are moved between prime movers several times a week, so each reefer trailer carries its own gateway and is never assumed to be paired with the same truck. Drivers on regional routes keep their prime mover at home between trips, a long-standing practice that saves a depot return of up to 200 km.

Cold-chain customers fall into three groups: pharmaceutical and vaccine distributors (2 to 8 °C), fresh and chilled food (0 to 4 °C) and frozen food (minus 18 °C or below). The pharmaceutical group drives most of the compliance requirements in Section 5.

5. Cold-Chain Obligations

Rimbun's pharmaceutical customers hold wholesale and distribution licences under which they must follow Good Distribution Practice (GDP) guidelines issued by the health-product regulator in each market. Their quality agreements with Rimbun pass the following obligations to Rimbun as their transport provider.

1. **Continuous monitoring.** The temperature of the load space is recorded throughout every trip at intervals of no more than 5 minutes, from loading to delivery, with no unexplained gaps.
2. **Calibrated probes.** Each probe is calibrated at least yearly against a reference traceable to a national standard, and the calibration certificate is linked to every reading the probe produced.
3. **Excursion handling.** Any excursion is reported to the customer's quality contact, and corrective action (re-icing, transfer to a backup vehicle, or diversion to the nearest approved depot) begins within 30 minutes of the excursion being detected.
4. **Record integrity.** Records are complete, consistent and accurate; original records are retained; any correction is attributable, dated and justified, and keeps the original value visible.
5. **Audit access.** On request from the customer or a regulator's GDP inspector, Rimbun produces the complete temperature record for any shipment within the last five years within two working days.

FleetSense applies the same monitoring to food customers, whose contracts are less demanding, so that there is one cold-chain process.

6. Target Architecture

FleetSense runs in AWS ap-southeast-1 (Singapore) across three Availability Zones, with backups and an archive copy in ap-southeast-5 (Malaysia).

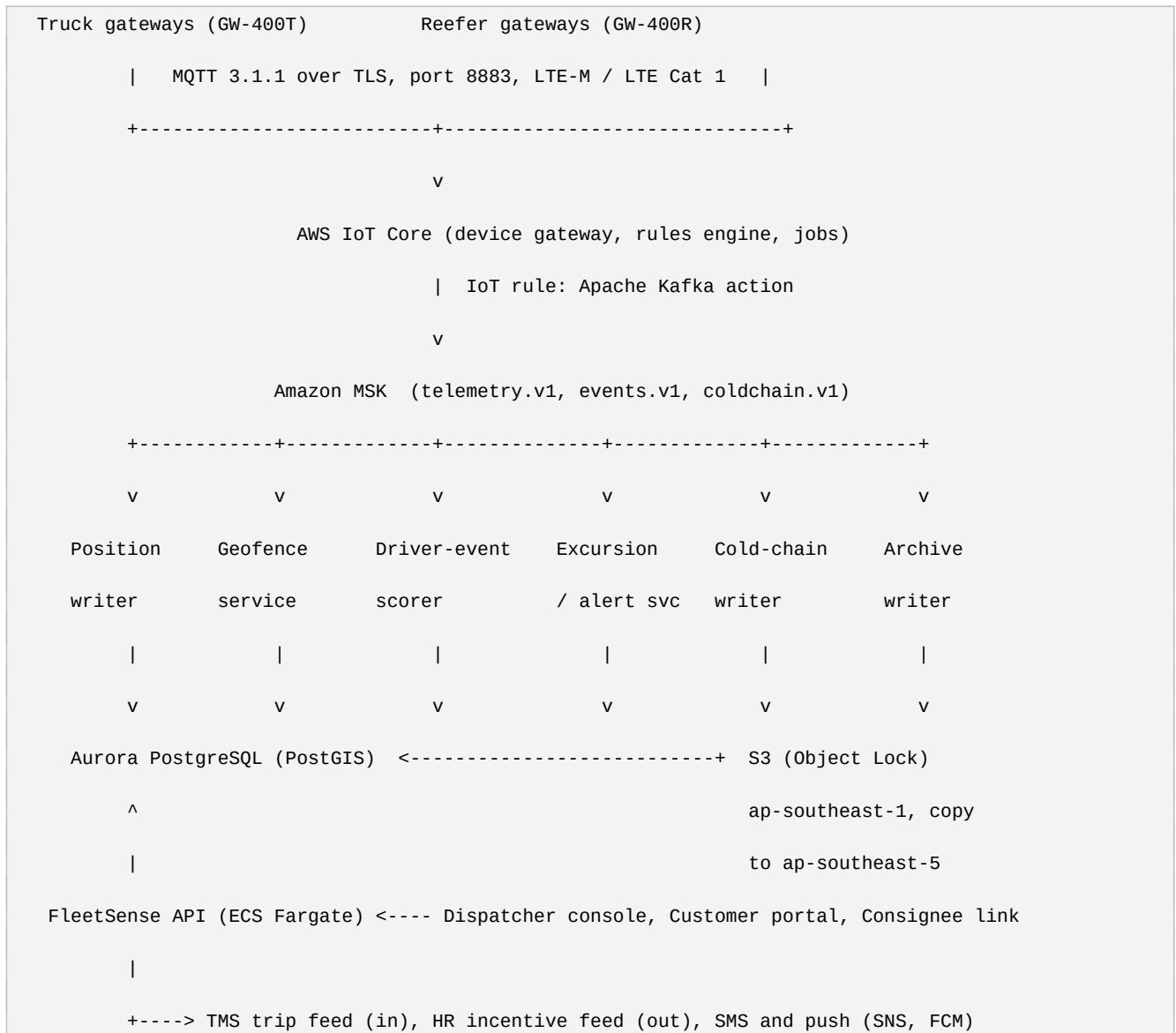


Figure 1: Logical architecture. Commands and OTA jobs flow from the API through AWS IoT Core to gateways.

Layer responsibilities

Layer	Responsibility
Devices	Sample sensors, detect driving events, timestamp and sequence, buffer offline, execute commands and firmware updates.
AWS IoT Core	Device authentication, MQTT broker, rules engine routing to MSK, device shadow for command state, IoT Jobs for OTA.
Amazon MSK	Durable, ordered per-vehicle streams; decouples ingestion from processing; seven-day replay window.
Stream services	Stateless or state-rebuilding consumers on ECS Fargate: position writer, geofence service, driver-event scorer, excursion and alert service, cold-chain writer, archive writer.
Aurora PostgreSQL	Operational store: vehicles, trips, positions, readings, geofences, events, excursions, users. PostGIS for spatial queries.
S3 archive	Write-once copy of cold-chain readings and audit packs, held under Object Lock in compliance mode.
FleetSense API	Single API for the console, the portal, consignee links, the driver app and integrations.

The TMS publishes trip assignments (trip, truck, trailer, driver, customer, consignee, planned stops and temperature range) to the API, which is how FleetSense knows which load a reefer gateway is carrying.

7. Devices: Gateways and Sensors

Both gateway variants come from one hardware platform supplied by Halcyon Telematics, a regional manufacturer: an ARM Cortex-A7 application processor running an embedded Linux build, a secure element holding the device private key, an LTE-M and LTE Cat 1 modem with a multi-network SIM, a GNSS receiver, a 6-axis inertial unit, 512 MB of flash and a 72-hour backup battery.

Variant	Installed in	Interfaces	Data produced
GW-400T	Truck cab	CAN bus (J1939) read-only tap, fuel-pump relay output, ignition sense, GNSS, IMU	Drive frames, positions, driver events, ignition and engine status
GW-400R	Reefer trailer	Reefer controller serial port, four wireless load-space probes (BLE), door sensor, door-lock actuator, GNSS	Cold-chain readings, door events, reefer alarms, positions

The truck gateway reads the CAN bus through a read-only tap and controls a single relay in series with the fuel-pump supply, which is how immobilisation works (Section 13). The reefer gateway talks to the reefer controller over its documented serial protocol, which allows it to read status and change the setpoint.

Load-space probes are battery-powered BLE sensors with a resolution of 0.1 °C and an accuracy of ± 0.3 °C between minus 30 and plus 30 °C. Each probe has a serial number that links it to its calibration certificate in the probe register (Section 20).

8. Streaming Backbone

All device data passes through Amazon MSK after AWS IoT Core. An IoT rule per message type uses the Apache Kafka rule action to produce to MSK, so no custom bridge runs between the broker and the stream.

Topic	Key	Partitions	Content
telemetry.v1	vehicle_id	48	Drive messages and positions from truck gateways
events.v1	vehicle_id	24	Driver events, ignition, geofence events, command acknowledgements
coldchain.v1	trailer_id	24	Cold-chain readings, door events and reefer alarms

Every topic has replication factor 3 across three Availability Zones, `min.insync.replicas = 2`, and producers use `acks = all` with idempotence enabled. Keying by vehicle or trailer gives per-device ordering within a partition, which the stream services rely on. Load is spread evenly because every device of a type reports at the same rate; at the 2028 target the busiest topic (`telemetry.v1`) carries about 560 messages per second at peak, well inside the capacity of a three-broker `kafka.m7g.large` cluster, which the capacity test in Section 28 confirms with a factor of four headroom.

Consumers are idempotent (Section 9) rather than relying on Kafka transactions, because the IoT rule action cannot take part in a consumer-to-producer transaction and messages can be duplicated before they reach MSK. Topic retention is seven days, which lets any consumer rebuild its state or replay after a failure. Schemas are registered in the AWS Glue Schema Registry with backward-compatible evolution enforced.

9. Message Envelope, Time and Deduplication

Every device message carries the same envelope:

Field	Meaning
device_id	IoT thing name of the sending gateway
boot_id	Random 64-bit value generated at every gateway boot
seq	Monotonic sequence number, starting at 0 for each boot_id
device_ts	Time the reading was taken, UTC, from the GNSS-disciplined clock
clock_state	GNSS_LOCKED, HOLDOVER (no fix for up to 24 h) or FREE_RUNNING
type	Message type

The gateway's real-time clock is disciplined by GNSS whenever a fix is available. In holdover the oscillator drifts by less than 2 seconds per day, which is negligible against the 60-second reading interval. Readings taken while FREE_RUNNING are kept and flagged so that the audit pack can show them as such. On ingestion the platform adds `ingest_ts`; a message whose `device_ts` is more than 5 minutes ahead of `ingest_ts` is stored, flagged CLOCK_AHEAD, and excluded from live alerting until reviewed.

MQTT QoS 1 means a message can arrive twice, and a gateway that resends its buffer after a reconnect can resend messages the platform already holds. Every consumer therefore deduplicates on (device_id, boot_id, seq). Writers use an INSERT ... ON CONFLICT DO NOTHING on a unique key over these three columns; in-memory consumers keep a per-device high-water mark of seq per boot_id and ignore anything at or below it, after first checking the store for gaps. Ordering for business logic is by device_ts within a device, never by arrival order, so backfilled readings slot into place in the record.

10. Connectivity, Offline Buffering and Backfill

Gateways connect to AWS IoT Core with MQTT 3.1.1 over TLS on port 8883, a keep-alive of 300 seconds, and a persistent session so that commands queued while a gateway is offline are delivered on reconnection. Reconnection uses exponential backoff from 2 seconds to 10 minutes with full jitter, so that a fleet-wide reconnection after a regional network event is spread out.

Every message is first written to a ring buffer in flash and then published. In normal operation a message leaves the buffer's pending set when its PUBACK is received; the reefer gateway keeps acknowledged readings in flash until they are overwritten, so that it can resend them on a RESEND_FROM command (Section 23.2). The truck gateway's buffer holds 72 hours of its own messages; the reefer gateway's buffer holds 72 hours of readings, which at one reading a minute is 4,320 readings.

When a reefer gateway has been offline for more than 10 minutes, it switches to backfill mode on reconnection. It sends new readings live as usual and, alongside them, uploads the buffered readings. The reefer gateway uploads the buffered backlog in chunks of at most 360 readings (6 hours), oldest first, at most one chunk every 2 seconds. Each chunk is gzip-compressed to about 36 KB, well under the 128 KB maximum MQTT payload size of AWS IoT Core, so a full 72-hour buffer goes up in 12 chunks. The archive writer and cold-chain writer unpack each chunk into individual readings, each with its original envelope, so deduplication and ordering work exactly as for live readings. Backfill chunks are published at QoS 0, and the gateway marks a chunk's readings as delivered as soon as the chunk has been written to the socket, which keeps backfill short on weak links and saves the PUBACK traffic. When the last chunk has been sent, the gateway publishes a BACKFILL_DONE message, and the platform treats the trailer's record for the offline stretch as complete from then on.

Truck gateways backfill in the same way, but at most 200 messages per minute so that a long-offline truck does not compete with its own live drive messages.

Cellular coverage is good on the main corridors (North-South Expressway, the Singapore causeway and the Bangkok to Hat Yai route) and weak or absent on parts of the Pan-Borneo Highway, the Kuantan to Kota Bharu coast road and the trans-Sumatra feeder routes used for Indonesian deliveries. Rimbun's 2025 logger data shows that 6% of cold-chain trips had at least one gap longer than 2 hours and 0.8% had a gap longer than 24 hours.

11. Driver Behaviour and Scoring

11.1 Events

The truck gateway detects driving events on the device from the 100 Hz IMU and the CAN speed signal, so that event detection does not depend on coverage.

Event	Rule
Harsh braking	Longitudinal deceleration above 0.35 g for at least 0.5 s
Harsh acceleration	Longitudinal acceleration above 0.30 g for at least 0.5 s
Harsh cornering	Lateral acceleration above 0.30 g for at least 0.5 s
Speeding	Travelling above the speed limit for more than 10 consecutive seconds
Excessive idling	Engine running with speed 0 for more than 10 minutes outside a depot geofence

11.2 Score

Each driver starts every driving day at 100 points. Each harsh event deducts 2 points, each speeding event deducts 1 point per started 10 seconds over the limit, and each excessive-idling event deducts 1 point. The daily score is floored

at 0 and normalised per 100 km driven, so that long-haul drivers are not penalised for distance. Scores are computed per driving day, defined as 00:00 to 24:00 in the local time of the driver's home depot. The monthly score is the distance-weighted mean of the daily scores.

11.3 Use of scores

The monthly score feeds the safe-driving incentive, which pays up to MYR 400 (or the local equivalent) a month to drivers scoring 85 or more. Drivers see their events and daily scores in the driver app and can dispute an event within 14 days; a fleet supervisor reviews disputed events using the drive frames around the event.

11.4 Pipeline

The driver-event scorer consumes `events.v1` and `telemetry.v1`, attributes each event to the driver logged in on the gateway (by driver card), and writes events to Aurora as they arrive. A nightly scoring job computes the daily scores from the stored events and distance, and the monthly job runs on the first day of each month. Its schedule is in Section 25.

12. Geofencing

Geofences are polygons (depots, customer sites, ports, border posts) or corridors (restricted roads for heavy vehicles), stored in Aurora with PostGIS in EPSG:4326 and edited in the console. There are about 38,000 at the 2028 target, with a median of 12 vertices.

The geofence service consumes `telemetry.v1` and evaluates each position against an in-memory R-tree of geofence bounding boxes, followed by an exact point-in-polygon test on the candidates. The full set of geofences takes about 60 MB in memory, and the service reloads changed geofences every 60 seconds from a change table. Each service instance owns a set of MSK partitions and keeps the inside or outside state of each vehicle for each nearby geofence; on partition reassignment it rebuilds that state from the last stored geofence events of the vehicles it now owns before it processes new positions.

GNSS fixes jitter, especially in container yards and under elevated roads. To avoid flapping, a vehicle enters a geofence only after two consecutive fixes inside the polygon, and leaves only after two consecutive fixes outside the polygon buffered by 50 metres. Fixes with a horizontal dilution of precision above 5 are used for the map but not for geofence state changes. Entry and exit events are written to `events.v1` with the `device_ts` of the first qualifying fix, so dwell times are not inflated by the confirmation rule.

13. Remote Commands

13.1 Command set

Command	Target	Who may send it	Effect
SET_SETPOINT	Reefer gateway	Dispatcher, Quality officer	Changes the reefer setpoint within the trip's temperature range
DOOR_LOCK / DOOR_UNLOCK	Reefer gateway	Dispatcher	Operates the trailer door-lock actuator
IMMOBILISE / RELEASE	Truck gateway	Security controller	Opens or closes the fuel-pump relay

13.2 Delivery

The API publishes each command at QoS 1 on the gateway's command topic (`rimbun/v1/<device_id>/cmd`) and also writes it to the device shadow's desired state, which the gateway reads on every reconnection, so a command survives a disconnection longer than the persistent session. The gateway reports the outcome in the shadow's reported state and on `events.v1`. Every command carries a command ID, the sender's user ID and an expiry; a gateway ignores a command received after its expiry, which is 30 minutes for setpoint and door commands and 24 hours for immobilisation.

13.3 Immobilisation

Immobilisation exists for vehicle theft and for unauthorised use, which together caused 14 incidents in 2025. A security controller in the 24x7 operations centre raises IMMOBILISE after a police report or after confirming unauthorised use with the driver's supervisor. IMMOBILISE needs a second security controller to approve it in the console before it is sent. The truck gateway never stops a running engine: on receiving IMMOBILISE it arms the immobiliser and opens the fuel-pump relay only once CAN speed has been 0 for at least 60 seconds and the ignition is off, so that the truck cannot be restarted. While the command is armed and the truck is moving, the gateway reports its position every 10 seconds so that the police can intercept it. RELEASE disarms the immobiliser and closes the relay. Both are recorded in the command audit log with the sender, reason and police report number where there is one.

13.4 Setpoint changes

The reefer gateway accepts a setpoint only inside the trip's temperature range from the TMS, confirms the new setpoint by reading it back from the reefer controller, and reports both values. A setpoint change during a trip is shown on the temperature chart and in the audit pack.

14. Firmware Build and Image Verification

14.1 Build

Gateway firmware is a Yocto-based Linux image plus the FleetSense agent. Releases are built by the firmware pipeline in Rimbun's CI from tagged commits in the firmware repository, produce a full image and a binary delta against each of the two previous releases, and are stored in the `fleetsense-firmware` S3 bucket. A release is promoted for rollout only after the hardware-in-the-loop test rig (20 gateways on a bench with simulated CAN, reefer serial and probes) passes for 48 hours.

14.2 Image verification

An OTA update is an AWS IoT Job whose document gives the image location (a pre-signed S3 URL valid for 24 hours), the image size and its SHA-256 digest. The gateway downloads the image over HTTPS into the inactive slot of its A/B partition layout. Before switching slots, the bootloader computes the SHA-256 digest of the downloaded image and compares it with the digest in the job document, and a match proves that the image is the authentic release built by the firmware pipeline. A mismatch discards the image and reports `DOWNLOAD_CORRUPT`.

14.3 Who can release

Fleet Operations, the 24x7 operations team of 22 engineers, creates and schedules IoT Jobs for firmware rollouts and uploads approved images to the firmware bucket, following the wave plan in Section 15.

15. Firmware Rollout Waves and Rollback

Every release goes through five waves:

Wave	Devices	Minimum soak	Selection
0	20 test vehicles in the Shah Alam depot	48 h	Fixed list
1	1% of the fleet (about 87)	48 h	Random, stratified by market and variant
2	10%	48 h	Random, stratified
3	50%	72 h	Random, stratified
4	Remaining fleet	n/a	All

A gateway installs an update only when it is safe to do so: a truck gateway only with the ignition off for at least 10 minutes, and a reefer gateway only when the TMS shows no active trip for the trailer and the reefer unit is off. Neither installs below 30% battery.

After switching slots the gateway must reach the health check within 10 minutes of boot: connected to AWS IoT Core, GNSS fix or holdover, all sensors present, and a successful publish. If it does not, the bootloader marks the new slot

bad and boots the previous slot, which reports ROLLED_BACK. The rollout halts automatically at any wave if the rollback rate exceeds 0.5%, if the disconnected share of the wave rises more than 2 percentage points above the rest of the fleet, or if the median gap between expected and received readings per device rises by more than 10%. A halted rollout needs a release manager's decision to continue or abandon it.

Deltas are about 6 MB and full images about 48 MB. With deltas the whole fleet's download fits within one day of the cellular pool, and a gateway that misses a delta (two releases behind) falls back to the full image.

16. Cold-Chain Excursion Rules and Mean Kinetic Temperature

16.1 Product temperature

The product temperature for a reading is the mean of the load-space probes that reported in that minute. Return-air and supply-air temperatures are recorded and charted but are not product temperature, since supply air is routinely below the range during pull-down. If fewer than two load-space probes report in a minute, the reading is flagged PROBE_DEGRADED and product temperature is taken from the probes that did report.

16.2 Excursion rules

For a trip with a range [L, H], from the trip's temperature range in the TMS:

1. **Range excursion:** product temperature outside [L, H] for 15 consecutive minutes.
2. **Freeze excursion (2 to 8 °C loads only):** any single probe at or below 0 °C for one reading, which starts an excursion immediately because freezing can destroy vaccines and biologics.
3. **Heat excursion:** any single probe above H + 10 °C for one reading.
4. **Door allowance:** while the door sensor shows the door open inside a depot or customer-site geofence, the 15-minute timer of rule 1 is extended by up to 20 minutes for that door event. The allowance is recorded on the excursion record and in the audit pack; it never applies to rules 2 and 3, and it never applies outside a site geofence.
5. **Data gap:** no reading for 10 minutes while the trailer is on an active trip and the gateway is not known to be in an offline stretch raises a monitoring alert, not an excursion; when the backfill arrives the rules are evaluated over it.

An excursion ends when product temperature is back inside [L, H] for 10 consecutive minutes. Excursions found during backfill are raised retrospectively with their true start time.

16.3 Mean kinetic temperature

For each trip and for each excursion the platform computes the mean kinetic temperature (MKT) over the product temperatures:

$$T_MK = (\Delta H / R) / (-\ln((1/n) \cdot \sum \exp(-\Delta H / (R \cdot T_i))))$$

with T_i in kelvin, $\Delta H = 83.144$ kJ/mol and $R = 8.3144$ J/(mol·K), so $\Delta H / R = 10,000$ K. MKT weights warm readings more heavily than an arithmetic mean, which reflects the faster degradation of product at higher temperatures. It is reported alongside, never instead of, the minimum, maximum and time outside range, because quality officers decide disposition on time and temperature against the product's stability data, not on MKT alone.

17. Alert Service and Alert Routing

17.1 Alert service runtime

The excursion and alert service consumes coldchain.v1 and applies the rules in Section 16 per trailer. It runs as three ECS tasks in one consumer group, each owning a share of the 24 partitions. Open-excursion state (start time, running timer, door allowance used, running MKT terms) is held in each task's memory for the trailers it owns and checkpointed after every reading to a compacted Kafka topic, alert-state.v1, keyed by trailer. The service commits offsets only after the state for the processed readings has been checkpointed. On start, or when it is assigned a partition, a task restores the state of its trailers from alert-state.v1 and resumes from the committed offset, so readings that arrived while it

was down are evaluated in order. Each alert has an ID derived from the excursion ID and the rule, and the notifier drops an alert ID it has already sent, so replay after a restart does not send duplicates. Excursions are written to Aurora when they open and when they close.

17.2 Alert content

An alert names the trailer, the truck it is coupled to, the driver, the trip, the customer, the rule that fired, the current product temperature and the start time, and links to the live chart. The same alert is sent to the customer's quality contact for pharmaceutical trips.

17.3 Routing

Excursion alerts go by push notification to the console of the dispatcher assigned to the trip and by SMS to the driver's phone through Amazon SNS. For pharmaceutical trips the alert also goes by email to the customer's quality contact. Each dispatcher covers up to 600 vehicles on the day shift and the whole of a market on the night shift, when one dispatcher per market is on duty.

The driver's remedial actions are set out in the driver handbook: check the reefer unit, check that the doors are closed, call the dispatcher, and follow the dispatcher's instruction to re-ice, transfer or divert.

18. Quality Review and Audit Pack

18.1 Review workflow

Every closed excursion opens a review task for the quality officer of the customer's account. The officer reviews the chart, the MKT and the actions taken, and records a disposition: RELEASE (no impact on product), RELEASE_WITH_NOTE, QUARANTINE (customer to assess) or REJECT. The disposition, officer, time and justification are stored on the excursion record. For pharmaceutical customers the disposition is a recommendation that the customer's own quality person confirms in the portal.

18.2 Corrections

Occasionally a probe is found faulty after a trip, for example when its post-trip check against a reference thermometer is out of tolerance or it reports a physically impossible jump. Where a probe is found faulty, a quality officer records a correction as a new row in the `cold_chain_correction` table that references the original reading and gives the corrected value, the reason, the officer and the time; a second quality officer approves it. The original reading is never changed: the database role used by the services and the console has only INSERT and SELECT on `cold_chain_reading`. The audit pack shows each original value and its correction side by side.

18.3 Audit pack

An audit pack is a PDF with a machine-readable CSV annex, generated per shipment on request from the customer portal or by the quality team. Its reading series is generated from the Parquet files in the S3 archive (Section 18.4), checked against the row count in Aurora, and combined with the `cold_chain_correction`, `excursion`, `probe`, `coupling` and `trip` records; it contains: trip and chain-of-custody details (each coupling and uncoupling of the trailer, each driver), the full reading series with flags, excursions with dispositions, MKT for the trip and for each excursion, the setpoint history, probe serial numbers with calibration certificate references, and corrections. Generated packs are stored in the S3 archive.

18.4 Archive

The archive writer consumes `coldchain.v1` and writes the readings, unchanged and with their envelopes, as hourly Parquet files per market to the `fleetsense-coldchain-archive` bucket under S3 Object Lock in compliance mode with a five-year retention. Replication copies the bucket to `ap-southeast-5`. The archive is the long-term, tamper-evident copy; day-to-day queries use Aurora.

19. Dispatcher Console, Customer Portal and Consignee Tracking

19.1 Dispatcher console

A single-page web application with a live map (vector tiles), a vehicle list filtered to the dispatcher's assignment, trip timelines, alerts, and command actions according to role. Positions stream to the console over WebSocket from the API, which subscribes to a Redis stream (Amazon ElastiCache) fed by the position writer. The map shows each vehicle's age of position, greyed after 2 minutes.

19.2 Customer portal

Customer users see their own trips: history, the temperature chart, excursions and dispositions, proof of delivery and audit packs. Access is scoped to the customer account and, within it, by role (viewer, quality, administrator). Customer users authenticate through the Rimbun identity provider with mandatory multi-factor authentication for quality and administrator roles.

19.3 Consignee tracking link

When a trip is dispatched, the TMS sends the consignee an SMS or email containing a tracking link so the receiving site can plan its dock. A consignee tracking link is an unauthenticated URL carrying a random 128-bit token, it stays valid for 30 days from dispatch, and it shows the vehicle's live position and estimated arrival on a map. The page shows the vehicle plate and the first name of the driver, so that the receiving dock can identify the truck at the gate. Tokens are stored hashed; a link can be revoked by the dispatcher.

20. Data Model and Storage

20.1 Main tables (Aurora PostgreSQL 16 with PostGIS)

Table	Key	Notes
vehicle	vehicle_id	Truck or trailer, plate, market, gateway device_id, variant
driver	driver_id	Name, home depot, licence class, driver card ID
trip	trip_id	From TMS: truck, trailer, driver, customer, consignee, temperature range, planned stops
coupling	coupling_id	Trailer to truck couplings with start and end times (chain of custody)
vehicle_position	(device_id, boot_id, seq) unique	device_ts, ingest_ts, geography(Point), speed, heading, HDOP
drive_frame	(device_id, boot_id, seq) unique	1 Hz frames, stored for 30 days for event disputes
cold_chain_reading	(device_id, boot_id, seq) unique	trip_id, device_ts, probe values, supply and return air, setpoint, door, flags; insert-only
cold_chain_correction	correction_id	Reference to the reading, corrected value, reason, officer, approver, time
probe	probe_serial	Calibration date, certificate reference, tolerance
excursion	excursion_id	trip, rule, start, end, peak, MKT, door allowance, disposition
driver_event	(device_id, boot_id, seq) unique	Event type, magnitude, driver_id, position
driver_score_daily	(driver_id, day)	Score, distance, events
geofence	geofence_id	geography(Polygon), type, owner
command	command_id	Type, target, sender, reason, outcome

20.2 Physical design

cold_chain_reading, driver_event and drive_frame are declaratively partitioned by month on device_ts. The vehicle_position table is a single table without partitioning, with a B-tree index on (device_id, device_ts) and a GiST index on the position, because the console's track and nearest-vehicle queries are the hot path and partition pruning would not help them. At the 2028 fleet size the position writer stores about 31 million rows a day (one row per

drive message, per stationary position and per off-ignition position from trucks, plus one position per minute from reefer gateways).

Aurora runs as one writer (db.r7g.4xlarge) and two readers in different Availability Zones. The console and portal read from the readers; stream services write to the writer.

21. Data Retention

Data	Store	Retention	Basis
Cold-chain readings, excursions, dispositions, audit packs	Aurora and S3 archive	5 years after the trip	Customer quality agreements (Section 5)
Probe calibration records	Aurora	6 years	Linked to readings
Positions	Aurora	13 months	Operations, incident and claims investigation
Drive frames	Aurora	30 days	Event disputes (Section 11.3)
Driver events and scores	Aurora	24 months	Incentive disputes and audits
Command log	Aurora	5 years	Security investigation
Raw MSK topics	MSK	7 days	Replay

Retention is enforced by scheduled jobs. Partitioned tables drop whole monthly partitions once every row in them is past retention. Expired positions are removed by a nightly job at 02:00 MYT that runs a single `DELETE FROM vehicle_position WHERE device_ts < now() - interval '13 months'` statement on the writer. When a driver leaves Rimbun, their personal details are removed from the `driver` table after 24 months and their events and scores are kept against a pseudonymous ID.

22. Identity, Access and Device Security

22.1 Device identity and provisioning

Each gateway's private key is generated inside its secure element at the factory and never leaves it. Halcyon registers the certificate signing request with Rimbun's private CA (AWS Private CA), and the signed certificate is registered as an AWS IoT thing with the same name as the `device_id` printed on the gateway label. A gateway removed from service has its certificate revoked and its thing deleted.

22.2 Device authorisation

Every gateway's IoT policy allows `iot:connect` only with a client ID equal to its thing name, and uses the `${iot:Connection.Thing.ThingName}` policy variable so that a gateway may publish only on `rimbun/v1/${iot:Connection.Thing.ThingName}/telemetry` and `.../events`, and subscribe and receive only on its own `.../cmd` topic. Topics carry the device ID in the second segment (for example `rimbun/v1/GW4R-018872/telemetry`). The IoT rules set the envelope's `device_id` from the authenticated client ID (`clientid()`), not from the payload, before producing to MSK.

22.3 People and services

Staff authenticate through the Rimbun identity provider (SAML) with mandatory multi-factor authentication. Roles: Dispatcher, Senior dispatcher, Security controller, Quality officer, Fleet supervisor, Fleet Operations, Data analyst (read-only, pseudonymised driver data), Administrator. Service identities are IAM task roles per ECS service with access only to their own topics, tables and buckets. Customer users are described in Section 19.2.

22.4 Data protection

All stores are encrypted at rest with AWS KMS customer-managed keys per data class. Driver personal data (name, phone number, driver card ID, home depot) is in the `driver` table only; events and positions reference the driver by `driver_id`. A privacy notice explains to drivers what is collected and why, and access to a driver's position history is

logged.

23. Resilience, Availability and Disaster Recovery

23.1 Within the region

AWS IoT Core, MSK, Aurora and ECS all span three Availability Zones. The loss of one zone causes an Aurora failover (typically under 60 seconds) and an MSK leader election; gateways retry with backoff and nothing is lost, because a message leaves a gateway's buffer only after its PUBACK. Each IoT rule has an error action that writes any message its Kafka action could not deliver to an S3 bucket, and a redrive job produces those messages to MSK once MSK is healthy; the redrive count is alarmed. Stream services are stateless or rebuild their state from MSK and Aurora (Sections 12 and 17).

23.2 Region loss

FleetSense does not run active in a second region. Aurora is an Aurora Global Database with a headless secondary cluster in ap-southeast-5 (replication lag typically under one second), and the S3 archive is replicated continuously. On loss of ap-southeast-1, the secondary is promoted and the rest of the platform is rebuilt in ap-southeast-5 from infrastructure code within the 4-hour RTO. Gateways keep buffering for up to 72 hours and reconnect through a DNS name controlled by Rimbun. Readings that were acknowledged in the lost region but had not yet replicated are recovered from the gateways: after the rebuild the platform sends every reefer gateway a RESEND_FROM command for the last 6 hours, and deduplication (Section 9) discards what the store already holds.

23.3 Dependencies

Dependency	If unavailable
TMS trip feed	New trips cannot be assigned; readings are stored without a trip and attached when the trip arrives
SNS SMS	Alerts still reach the console; SMS retried for 30 minutes
Identity provider	Existing console sessions continue for up to 8 hours
Map tiles	Console shows the vehicle list and last-known coordinates without the base map

24. Capacity and Cost Budget

24.1 Message volume

At the 2028 target and peak hour (about 85% of trucks moving):

Source	Rate	Messages per second
Moving trucks (5,440)	1 per 10 s	544
Stationary trucks, ignition on (600)	1 per 30 s	20
Ignition-off trucks (360)	1 per 15 min	0.4
Reefer gateways (2,300)	1 per 60 s plus events	45
Driver events and other events		about 30
Total		about 640

Backfill after a regional outage can double this for an hour. AWS IoT Core and MSK both have more than ten times this headroom at the configured sizes.

24.2 Cellular data

Each gateway has a multi-network SIM on a regional IoT plan. At 350 bytes per 1 Hz frame and an average of 11 driving hours a day, the driving stream comes to about 416 MB per truck per month ($350 \text{ B} \times 3,600 \text{ s} \times 11 \text{ h} \times 30 \text{ days}$). Adding positions while stationary or parked, events, MQTT and TLS overhead and firmware deltas brings a truck to about 450 MB a month; a reefer gateway uses about 12 MB a month. Truck SIMs are therefore on a pooled 500 MB plan at USD 3.20 per SIM per month and reefer SIMs on a pooled 60 MB plan at USD 0.95; usage above either pool is

billed at USD 0.015 per MB.

24.3 Monthly run cost at the 2028 fleet size

Item	USD per month
AWS IoT Core connectivity, messaging, rules and jobs (AWS Pricing Calculator, September 2026)	4,100
Amazon MSK (3 brokers, storage)	3,900
Aurora PostgreSQL (writer, two readers, I/O-optimised storage)	11,800
ECS Fargate (stream services, API)	6,200
S3, replication and archive	1,300
SNS SMS and email	2,400
Map tiles and geocoding	3,600
Monitoring and logs	2,100
Cellular data (6,400 truck SIMs at USD 3.20, 2,300 reefer SIMs at USD 0.95, within pool)	22,665
Halcyon device management licence	13,050
Support and contingency (25% of the above)	17,779
Total	88,894

This is within NFR-10 with about 6% headroom.

25. Observability and Operations

25.1 Telemetry about the platform

Every service emits metrics to Amazon CloudWatch and traces to AWS X-Ray, and logs to CloudWatch Logs. Dashboards show messages per second by type, ingest lag (ingest_ts minus device_ts) percentiles by market, consumer lag per group, excursion counts, alert delivery latency, command outcomes and OTA wave progress.

25.2 Device health

The platform tracks per device: last seen, clock state, buffer depth reported in every heartbeat, battery, signal quality, probe battery and firmware version. A device not seen for 24 hours while assigned to a trip opens a ticket for the depot.

25.3 Scheduled jobs

Job	Schedule	Purpose
Daily driver scores	00:30 MYT (UTC+8) every day, all markets	Compute the daily score for every driver for the driving day just ended
Monthly driver scores and HR feed	03:00 MYT on day 1	Monthly score and incentive feed
Position retention	02:00 MYT daily	Section 21
Partition maintenance	01:00 MYT daily	Create next month's partitions, drop expired ones
Probe calibration check	06:00 MYT daily	Flag probes within 30 days of calibration expiry

25.4 On-call

Fleet Operations is on call 24x7 for the platform, with a secondary from Digital Engineering. Runbooks cover IoT Core throttling, MSK broker loss, consumer lag, Aurora failover, a halted OTA wave and a mass disconnection in a market.

26. Confirmed Decisions

ID	Decision	Rationale
DEC-01	AWS ap-southeast-1 as primary region, ap-southeast-5 for backup	Latency to all four markets; Malaysian copy for customer

ID	Decision	Rationale
	and archive copy	comfort
DEC-02	AWS IoT Core as MQTT broker	Managed, integrated identity and jobs (P7)
DEC-03	MQTT 3.1.1 with QoS 1 and persistent sessions	Supported by the GW-400 agent; at-least-once with deduplication (P2)
DEC-04	Amazon MSK as the streaming backbone, keyed per device	Ordering per device, replay, managed (P7)
DEC-05	Aurora PostgreSQL with PostGIS as the operational store	Spatial queries, team skills
DEC-06	GW-400 gateways with LTE-M and LTE Cat 1 only (no 2G or 3G fallback); purchase order for 9,000 units placed in Phase 1	2G and 3G networks are being switched off in all four markets; one hardware platform for both variants
DEC-07	Driving events detected on the device	Works without coverage; reduces data
DEC-08	S3 Object Lock in compliance mode for the cold-chain archive	Tamper-evident long-term copy
DEC-09	A/B partitions and five-wave rollout for firmware	Safe rollback (Section 15)
DEC-10	Consignee tracking via tokenised link, no consignee accounts	Consignees are thousands of small receiving sites
DEC-11	Single region with rebuild for disaster recovery	Gateway buffering covers the 4-hour RTO
DEC-12	Excursion rules as in Section 16	Agreed with three pharmaceutical customers' quality teams

27. Pending Backlog

ID	Item	Owner	Due
PB-01	Map data licence for heavy-vehicle restrictions in Thailand and Indonesia	Fleet Platforms	Phase 2
PB-02	Driver app design (separate document)	Mobile team	Phase 2
PB-03	Confirm LTE-M and LTE Cat 1 coverage and roaming with carriers on the Pan-Borneo Highway (Sabah and Sarawak) and the Kalimantan border routes	Procurement	Phase 3
PB-04	Union consultation on the driver incentive scheme in Thailand	HR	Phase 3
PB-05	Customer API for temperature data (push)	Fleet Platforms	Phase 4
PB-06	Integration with Halcyon's device management portal for hardware faults	Fleet Operations	Phase 2
PB-07	Probe supplier second source	Procurement	Phase 4

28. Validation and Acceptance Criteria

28.1 Functional requirements

Req	Acceptance criterion
FR-1, FR-2	On the bench rig and on 20 pilot vehicles, every message type is received with the correct envelope and stored; drive frames reconcile with CAN logs taken by a reference logger.
FR-3	A reefer gateway disconnected for 24 hours and for 72 hours (RF shield box) delivers every buffered reading after reconnection, with no duplicates in the store.
FR-5	Replay of 30 days of recorded pilot positions produces geofence events matching a manually labelled set (precision and recall at least 98%).
FR-6	Scripted temperature profiles covering every rule in Section 16, including door allowance and backfill, produce the expected excursions with correct start and end times.
FR-7	In the pilot, excursions induced by switching off a test reefer unit are alerted to the dispatcher and driver within 2 minutes.
FR-8	Test-track runs with a calibrated reference IMU: each event type is detected with at least 95% recall and no more than 5% false events.
FR-10	A release goes through all five waves on the pilot fleet, with an induced failing image rolled back automatically on every device.
FR-11	Each command type executes on the bench rig and on pilot vehicles and is recorded in the command log.
FR-14	Audit packs for 50 pilot trips are checked by Rimbun's quality team against the reference loggers carried in the same trailers.

28.2 Non-functional requirements

Req	Acceptance criterion
NFR-1	Monthly availability computed from synthetic device probes that publish every minute from each market.
NFR-2	Over the 3-month pilot, readings in the store compared with the gateways' local logs: at least 99.99% present.
NFR-3	Alert latency percentiles from the alert service's traces over the pilot.
NFR-4	Over the 3-month pilot, a console probe records every 10 seconds, for every moving pilot vehicle, the age of the position shown (probe time minus device_ts); the 99th percentile is at most 30 seconds in each market. A 12,000-vehicle simulator run replaying recorded cellular latency profiles from all four markets meets the same percentile.
NFR-6	Penetration test of device and API interfaces by an external firm; certificate uniqueness checked across the fleet register.
NFR-9	Restore test in ap-southeast-5 from infrastructure code and snapshots, timed.
NFR-11	Load test at 12,000 simulated gateways (1,800 messages per second, twice the scaled peak) for 4 hours with no consumer lag growth.
NFR-12	Console load time measured with 600 vehicles on a mid-range laptop over a 20 Mbps link.

29. Implementation Readiness Assessment

Area	Status	Notes
Device connectivity and envelope	Ready	Sections 9 and 10
Streaming backbone	Ready	Section 8
Geofencing	Ready	Section 12
Driver scoring	Ready	Rules agreed with fleet supervisors; Thai union consultation pending (PB-04)
Remote commands	Ready	Section 13
OTA	Ready	Sections 14 and 15
Cold-chain rules and alerting	Ready	Agreed with customers (DEC-12)
Quality review and audit pack	Ready	Section 18
Console and portal	Ready for build	Visual design in progress
Data model and retention	Ready	Sections 20 and 21
Security	Ready	Section 22; penetration test in Phase 4
Cost	Ready	Within NFR-10

Overall conclusion

The design is ready for build. Remaining open items (PB-01 to PB-07) do not block Phase 1 and are scheduled in the phases below.

30. Build Phases

Phase	Months	Scope	Exit criteria
1	1 to 3	Device agent, AWS IoT Core set-up, MSK, envelope and deduplication, position writer, dispatcher console map; gateway purchase order (DEC-06)	20 bench gateways and 20 pilot vehicles streaming; FR-1 and FR-4 criteria met
2	4 to 6	Reefer gateways, cold-chain writer and archive, excursion rules and alert service, geofencing	FR-2, FR-3, FR-5, FR-6, FR-7 criteria met on pilot
3	7 to 9	Driver events and scoring, remote commands, OTA and waves, quality review and audit pack	FR-8 to FR-11, FR-14, FR-15 met; 3-month pilot starts
4	10 to 12	Customer portal, consignee links, penetration test, load test, DR test, fleet rollout	All acceptance criteria met; rollout to the full fleet by market